Exercise Set 3

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In the lecture, you have seen Coppersmith's method for finding small roots of $F(x) \equiv 0 \pmod{M}$: polynomials are turned into vectors, LLL finds a short vector, and Howgrave-Graham's theorem turns it back into a polynomial with the root over the integers. Problem 1 makes the correspondence between polynomials and vectors precise. Problem 2 extends the method to roots modulo an unknown divisor of N, i.e. the setting of the motivating example of the lectures where the modulus is the unknown p and only $N = pq$ is available.

Problem 1 (Polynomial-vector transformations). Denote by $\mathbb{Z}[x]$ the ring of polynomials with integer coefficients and fix a positive integer X. In the lectures, we have defined a way to convert between polynomials of degree at most d in $\mathbb{Z}[x]$ and (row) vectors in $\mathbb{Z}^{d+1}$.

Consider the following function:

$$\begin{aligned} \mathsf{poly2vec}: \quad & \mathbb{Z}[x] \to \mathbb{Z}^{d+1} \\ & a_0 + a_1 x + a_2 x^2 + \ldots + a_d x^d \mapsto [a_0, a_1 X, a_2 X^2, \ldots, a_d X^d] \end{aligned}$$

Let $\mathsf{vec2poly}$ be the following operation:

$$\begin{aligned} \mathsf{vec2poly}: \quad & \mathbb{Z}^{d+1} \to \mathbb{Z}[x] \\ & [v_0, v_1, v_2, \ldots, v_d] \mapsto v_0 + \frac{v_1}{X} x + \frac{v_2}{X^2} x^2 + \ldots + \frac{v_d}{X^d} x^d \end{aligned}$$

(a) Consider a polynomial $F(x) \in \mathbb{Z}[x]$. Prove that the result of $\mathsf{poly2vec}(F)$ can be computed by taking the coefficient vector of $F(xX)$ (i.e. the polynomial F evaluated at $x \cdot X$).

(b) Argue that the function $\mathsf{vec2poly}$ is well defined. That is, prove that, for any polynomial $F(x)$ of degree at most d, $\mathsf{vec2poly}(\mathsf{poly2vec}(F)) = F$ and that all operations are well defined.

(c) Consider two polynomials $F_1(x), F_2(x) \in \mathbb{Z}[x]$ of degree at most d with a common root x_0 and their corresponding vectors $\mathbf{v}_i := \mathsf{poly2vec}(F_i)$, for $i \in \{1, 2\}$. Prove that the following operations are well-defined and their results are polynomials that also have a root at x_0.

- $\mathsf{vec2poly}(\mathsf{poly2vec}(F_1) + \mathsf{poly2vec}(F_2))$
- $\mathsf{vec2poly}(k \cdot \mathsf{poly2vec}(F_1))$ for $k \in \mathbb{Z}$

(d) Consider $\mathbb{Z}_N$, where N is an RSA modulus. Assume we use Coppersmith's method to find a small root x_0 of a polynomial $F(x) \in \mathbb{Z}_N[x]$. Argue that the matrix *after* the application of LLL consists of rows, all of which encode polynomials that have x_0 as a root modulo N.

Problem 2 (Generalizing Coppersmith's Method). In the lecture you have seen that Coppersmith's method can be used to find small roots of a polynomial $F(x)$ modulo N, even when the factorization of N is not known. In this exercise, we show that Coppersmith's method can also be employed to find small roots of a polynomial $F(x)$ *modulo a divisor of N, even if we do not know the divisor*. This, however, requires changing the algorithm a bit.

Let N be a large integer of unknown factorization.[1] Let b be an unknown divisor of N with $b \geq N^\beta$ for some known $0 < \beta \leq 1$.[2] Let $F(x) \in \mathbb{Z}[x]$ be a monic polynomial of degree d with $F(x_0) \equiv 0 \pmod{b}$ for some x_0 with $|x_0| < X$, $X \in \mathbb{Z}$. Let $h \geq 2$ be an integer to be fixed later.

[1] Recall that, if the factorization is known, one can split the problem into sub-problems, one for each factor. Each sub-problem consists of finding a root modulo a prime p, which is easy. One can then compose the final solution by using the Chinese Remainder Theorem.

[2] Note that the case $\beta = 1$ corresponds to the case already analyzed in the lectures.

Consider the sequence of polynomials from the "Full Coppersmith Method" from the lectures:

$$G_{i,j}(x) = N^{h-1-j}F(x)^j x^i, \quad 0 \le i < d,\ 0 \le j < h$$

Note that all $G_{i,j}(x_0) \equiv 0 \pmod{b^{h-1}}$ and that $G_{i,j}$ has degree $dj + i$. Let $\mathcal{L}$ be the lattice with $\mathsf{poly2vec}(G_{i,j})$ as basis vectors. Note that $\mathcal{L}$ has dimension dh.[3]

(a) Consider $F(x) = x + a$. Assume that $F(x_0) \equiv 0 \pmod{b}$, where $b \ge N^{\frac{1}{2}}$ (thus $\beta = 1/2$) and $|x_0| < X$ for some $X \in \mathbb{Z}$. Show that the method above cannot provably find this small root x_0, even for large values of h (e.g. $h = 50$).

Hint: this requires you to show that there does not exist any value of X that satisfies the Howgrave-Graham condition.

(b) Let $t = \lfloor dh(1/\beta - 1) \rfloor$. Consider now the following additional polynomials:

$$H_i(x) = x^i F(x)^h, \quad 0 \le i < t$$

Once again, note how $H_i(x_0) \equiv 0 \pmod{b^{h-1}}$. Show that for $h = 2$, including both polynomials $G_{i,j}$ and H_i, we can provably find the root x_0 for the polynomial above if $X < N^{\frac{1}{6}}/2^{\frac{7}{6}}$.

[3]The maximum degree of any polynomial is $d \cdot (h-1) + (d-1) = dh - 1$, which means that there are $dh - 1 + 1 = dh$ coefficients